



Conversion of organic material by black soldier fly larvae: establishing optimal feeding rates

Stefan Diener, Christian Zurbrügg, Klement Tockner*

Swiss Federal Institute of Aquatic Science and Technology, Dübendorf, Switzerland

Larvae of the black soldier fly, *Hermetia illucens* (Diptera: Stratiomyidae), are voracious feeders of organic material and may thus be used in simple engineered systems to reduce organic waste in low- and middle-income countries. Controlled feeding experiments with standard fodder were conducted to assess the optimum amount of organic waste to be added to a CORS system (Conversion of Organic Refuse by Saprophages). A daily feeding rate of 100 mg chicken feed (60% moisture content) per larva resulted in an optimum trade-off between material reduction efficiency (41.8%, SE 0.61) and biomass production (prepupal dry weight: 48.0 mg, SE 2.0). Applied to market waste and human faeces, this corresponds to a potential daily feeding capacity of 3–5 kg/m² and 6.5 kg/m², respectively. In addition, *H. illucens* prepupae quality was assessed to determine their suitability to substitute fishmeal in animal feed production. The chitin-corrected crude protein content ranged from 28.2 to 42.5%, depending on the amount of food provided to the larvae. Based on our study, a waste processing unit could yield a daily prepupal biomass of 145 g (dry mass) per m². We conclude that larvae of the black soldier fly are potentially capable of converting large amounts of organic waste into protein-rich biomass to substitute fishmeal, thereby contributing to sustainable aquaculture.

Keywords: *Hermetia illucens*, larval development, organic waste treatment, conversion of organic refuse by saprophages (CORS), aquaculture, protein production

Introduction

Low waste collection coverage and inadequate treatment of waste lead to unbearable conditions in villages and cities of low- and middle-income countries. A global survey by the United Nations Development Program (UNDP) in 151 cities revealed that poor solid waste disposal faced by city dwellers was considered the second most serious problem after unemployment (UNDP 1997). It impacts human health, the environment, and is also a major obstacle to economic development. In contrast, well functioning informal waste management systems are inherently linked to job opportunities and income generation. Specifically, recycling of inorganic material (e.g. glass, metal, plastics) from municipal solid waste is already an important source of income and employment of the informal sector, especially in low- and middle-income countries (e.g. Ali 1999, WASTE & Skat 2006). However, one- to two-thirds of the solid waste is not collected in developing countries. In particular, organic waste reuse is still in its embryonic stage despite its high recovery potential. Organic material can make up more than 50% of the total municipal waste pro-

duction (e.g., Ojeda-Benitez *et al.* 2003, Henry *et al.* 2006, Sharholly *et al.* 2006). For lack of alternatives, solid waste as well as human and animal excreta are often dumped arbitrarily in streets and drains. Heavy rainfall may clog the drains and lead to urban and suburban flooding. Furthermore, organic waste deposits will not only form breeding places for pathogen-transmitting insects and rodents, but also cause odour nuisance. Given the large fraction of organic waste, simple and efficient collection and organic waste treatment systems may significantly mitigate these problems and enhance recycling of organic matter and nutrients.

Conversion of organic refuse by Saprophages (CORS) has been an issue over the last few decades (Beard & Sands 1973, El Boushy 1991, Barnard *et al.* 1998, Ramos-Elorduy *et al.* 2002, Elissen *et al.* 2006). The most familiar example of CORS is vermicomposting, namely the conversion of organic waste by worms and micro-organisms into black, earthy-smelling, nutrient-rich humus. Similarly, larvae of *Hermetia illucens* L. (Diptera: Stratiomyidae), better known as the black soldier

Corresponding author: Stefan Diener, Swiss Federal Institute of Aquatic Science and Technology, Überlandstrasse 133, 8600 Dübendorf, Switzerland.
E-mail: Stefan.Diener@eawag.ch

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*This author is also affiliated to: Leibnitz-Institute of Freshwater Ecology and Inland Fisheries, Berlin and Institute of Biology, Free University Berlin, Berlin, Germany.

fly, have been propagated as converters of organic waste, but also as a nutritious feed for chicken, pig breeding and aquaculture (Booram *et al.* 1977, Newton *et al.* 1977, Bondari & Sheppard 1981, Lardé 1990, Sheppard *et al.* 1994, Newby 1997). This non-pest fly originates from the southern USA and has spread throughout the tropics and subtropics. Its larvae feed on organic waste; however, since the adult fly does not take up any food (Sheppard *et al.* 2002), surviving only on its reserves, it does not pose any disease transmission risks. Mating takes place two days after emergence, and oviposition occurs two days after fertilization (Tomberlin & Sheppard 2002). Larval development requires 2–4 weeks, depending on temperature and food availability (May 1961, Schremmer 1986, Myers *et al.* 2008). The prepupae crawl out of the organic material in search of a dry pupation site. During this dispersal phase, prepupae can easily be harvested by simply constricting their dispersion paths (Sheppard 2002, Tomberlin *et al.* 2002). The prepupae are protein-rich, similar to commercial fishmeal, and may therefore be used as animal feed.

Previous studies focused on the degradation of cow, chicken or pig manure by *Hermetia illucens* larvae in cost- and maintenance-intensive systems of high-income countries (Sheppard *et al.* 1994, Newton *et al.* 2005a, b). However, little information is available on the capability of the larvae to digest municipal organic solid waste (MOSW) or to process agricultural waste (e.g. coffee, banana), as well as excreta and faecal sludge from on-site sanitation facilities (e.g. pit latrines) – all waste products whose disposal and management cause serious challenges in low and middle-income countries. Newby (1997) investigated the activity of *H. illucens* in a compost-type system at household level in Australia. Lardé (1990) tested the conversion of coffee pulp in a small-scale experiment and Hem (Hem *et al.* 2008) built up a low-tech system to treat palm kernel meal in the Republic of Guinea. Bradley (1930) already noted in 1930 that *H. illucens*-infested privies (i.e. pit latrines), thus indicating their potential to treat faeces. Although all the aforementioned studies showed promising results, the further development of this technology from an experimental scale into a practicable full-scale plant has not yet been tested.

In fact, organic waste continues to constitute tremendous problems in low- and middle-income countries, as no valid solution has yet been identified. Development from experimental to full-scale waste treatment facilities, using the larvae of the black soldier fly, offers several advantages. Since such facilities can be developed and operated at low cost (low building and maintenance costs; independent from power supply), they are more adapted to the economic potential of developing countries. Furthermore, creating additional value chains and generating a surplus income through the sale of harvested prepupae or their use in animal husbandry can strengthen the economic resilience of farmers or small entrepreneurs to natural hazards or market fluctuations. Agricultural studies in Africa revealed that the high cost of feed is an important factor for smallholder poultry production (Sonaiya 1993). Strategies to counteract high feed prices could consist in switching to other poultry species such as waterfowls, which may be raised with other feed (snails, water hyacinths). Another option would be to supplement the feed with alternative materials produced locally or even by the farmers themselves to ease the financial burden of the smallholder. Prein & Ahmed (2000) pointed out the advantages of integrated agriculture-aquaculture (IAA) systems by providing successful examples from Africa (Malawi, Ghana) and Asia (Bangladesh, Philippines). Ahmed & Lorica (2002) described the positive effects of small-scale aquaculture on household income, employment and consumption. Therefore, the use of a black soldier fly CORS system of appropriate design and feasible operation can meet the requirements of such extensive cultures, as the yield in prepupae can be used directly. In low and middle-income countries, the fly can act as an ecological engineer.

The high protein and fat content of dried soldier fly prepupae reinforces its high potential as fly meal in animal feed production (Table 1). A very attractive market for dried soldier fly prepupae is the rapidly growing aquaculture industry. Between 2002 and 2004, this economic activity grew worldwide by 6.1% on average. Some low- and middle-income countries such as Chile, Iran or Vietnam even showed growth rates as high as 11.2, 16.5 and 30.6%, respectively. Myanmar exhibited the highest growth rate with 40.1% (FAO 2007). Fishmeal and fish oil are the main food sources for farmed aquatic species.

Table 1: Relative contribution of protein and crude fat (% on a d.w. basis) to the composition of different insect meals.

	Protein (%)	Crude fat (%)	Source
Insect meal			
Black Soldier Fly, <i>Hermetia illucens</i>			
prepupa	44	33	(St-Hilaire <i>et al.</i> 2007b)
larva	42–45	31–35	(Booram <i>et al.</i> 1977, Hale 1973)
House fly, <i>Musca domestica</i>			
pupa	63	15	(Calvert <i>et al.</i> 1969, Ravindran & Blair 1993)
larva	38	20	(Ogunji <i>et al.</i> 2007)
Mealworm, <i>Tenebrio molitor</i>	48–58	29–38	(Ng <i>et al.</i> 2001; Ramos-Elorduy <i>et al.</i> 2002)
Fishmeal	62–70	8.9–9.3	(Sauvant <i>et al.</i> 2004)
Soybean meal	43–47	1.5–1.9	(Sauvant <i>et al.</i> 2004)

The rapid spread of aquaculture leads not only to an increase in demand for fishmeal derived from wild fish stocks, but to a rise in fishmeal price and pressure on the natural fish populations. Therefore, alternative protein sources, ideally of animal origin, may turn out to be a highly attractive alternative for farmers currently depending on fishmeal. The prepupae of *Hermetia illucens* may serve as such an alternative protein source.

A CORS system combining waste treatment with concurrent protein production has to meet two objectives: (1) to achieve a high waste reduction efficiency; and (2) to yield a maximum prepupal biomass. Therefore, it is necessary to determine the maximum amount of waste that can be processed by larvae per day. Waste is considered fully processed when it has been subject to one larval digestion cycle. The resulting residues can be used as nutritious soil amendment in agriculture. Digestion stabilizes the waste, reduces odour emissions as well as fungal and bacterial growth. In addition, for a CORS system to be economically viable, it must generate prepupae of attractive quality and quantity.

This paper experimentally tests the hypothesis that there is a threshold in food supply, where larval dry weight does not increase further and larval development time is no longer reduced by an increase in food supply. The experiments conducted identify the maximum amount of waste larvae can cope with within their development period.

The study addressed the following specific research questions.

1. What is the daily maximum food quantity (i.e. organic waste) digestible by *Hermetia illucens*? The highest overall material reduction rate in relation to larval development time was determined.
2. What is the daily optimum food quantity for black soldier fly prepupae to reach maximum dry mass within the shortest development time?
3. Does the protein content of prepupae differ with varying feeding rates?

Material and methods

Black soldier fly *Hermetia illucens* L. (Diptera: Stratiomyidae) larvae were obtained from a colony maintained at constant temperature (27 °C, 67% relative humidity) in a container (3 m × 2 m) in Dübendorf, Switzerland. The container was fitted with two windows, as direct sunlight is required for successful mating (Tomberlin & Sheppard 2002). Females oviposited in the flutes of strips of corrugated cardboard which were attached to the wall of a bucket containing wet chicken feed as an attractant (as proposed by Tomberlin & Sheppard 2002). The larvae were reared on chicken feed (UFA 625) thoroughly mixed with water (60% moisture). All feeding experiments were conducted in a temperature-controlled, insulated container ($T = 26$ °C (SE: 0.02), relative humidity = 67% (SE 0.14), 12 h photoperiod).

Each treatment (three replicates per treatment) contained 200 larvae (6 days old, hand counted) fed with five different daily food rates: 12.5, 25, 50, 100, and 200 mg larva⁻¹ day⁻¹

(wet weight, 60% moisture content, chicken feed UFA 625, apparent metabolizable energy for poultry = 11.7 MJ kg⁻¹), and are henceforth referred to as class 12.5, 25, 50, 100, and 200, respectively. The 6-day old larvae were initially placed onto the prepared food within plastic boxes (14 cm × 7.5 cm × 7 cm) and covered by black cardboard to protect them from light disturbance. Box lids contained nine holes (diameter, 15 mm) to allow air circulation. To avoid oviposition by other flies in the experimental box, mosquito mesh was clamped between the box and lid. Food rations were prepared, weighed, and kept frozen 24 h before use.

Sampling and feeding were performed every two to three days. Added food amounts were calculated according to the daily food rates mentioned above and the number of days until the next feeding. While sampling and feeding, larvae were transferred into another box containing the next food ration. Residual material of the previous box was dried for at least 24 h at 105 °C to determine its dry mass. At each transfer, five randomly selected larvae were removed for further analysis. The selected larvae were weighed (wet mass), lyophilized to measure dry weight and ground for nitrogen analysis (CHNS-O, EA 1108-Elemental Analyser, Carlo Erba). Assuming proteins from animal tissue to contain about 16% nitrogen (Jones 1931), crude protein content was calculated by multiplying the total nitrogen content with the conversion factor 6.25. However, as the nitrogen content in chitin amounts to 6.89%, such a conversion may lead to an overestimation of the calculated protein content. To take this factor into account, the chitin content of the prepupae was determined using the formic acid method (Lovell *et al.* 1968). The chitin-derived nitrogen content of the prepupae was subtracted from the total nitrogen content before protein calculation.

Prepupae, identified by means of their dark colour, were removed and placed into a box containing wood shavings to allow pupation in a dry environment. Feeding of larvae was continued until more than 50% of the larvae in a box had developed into prepupae. Five prepupae were sampled from each box and treated the same way as the larvae (see above).

The emerging adults were killed using ethyl acetate. Sex was determined and the individuals were lyophilized to assess their dry weight.

To take into account not only the overall material reduction but also the time the larvae require to reduce this amount of food, a waste reduction index (WRI, equation (1)) was calculated using the overall degradation D (equation (2)) divided by the number of days the larvae fed on the material.

$$WRI = \frac{D}{t} \times 100 \quad (1)$$

$$D = \frac{W - R}{W} \quad (2)$$

where W is the total amount of organic material applied during the time t and R is the residue after time t . The factor 100

Table 2: Insect growth dynamics and estimation of substrate utilisation.

	12.5 mg		25 mg		50 mg		100 mg		200 mg	
	Mean	SE	Mean	SE	Mean	SE	Mean	SE	Mean	SE
Development time (egg to prepupa, days)	42.2a	0.32	32.7b	0.66	20.1c	0.14	16.6d	0.31	15.9d	0.02
Feeding duration (days)	36.2	0.19	26.7	0.18	14.1	0.07	10.6	0.07	9.9	0.05
Substrate consumption (%)	39.7ab	1.10	37.3b	0.55	43.2a	1.41	41.8ab	0.61	26.2c	1.40
WRI (Waste Reduction Index, substrate consumption/feeding time)	1.1a	0.03	1.3a	0.02	3.1b	0.10	3.8c	0.06	2.4d	0.13
ECD (Efficiency of Conversion of Digested-Feed, %)	38.0a	0.60	33.3a	1.40	26.2b	2.10	24.4b	0.43	25.8b	1.00

Daily feeding rates: 12.5, 25, 50, 100 and 200 mg layer hen feed (60% moisture) per larva and day.

Mean values followed by the same letter in the same row do not vary significantly ($P < 0.05$).

is used to give the index a practical value. High *WRI* values indicate good reduction efficiency.

Quantitative nutritional aspects are based on the terminology of Scriber & Slansky (1981): $B = (I - F) - M$ and $ECD = B/(I - F)$, where B is the assimilated food used for growth (measured as prepupal biomass), I is the total food offered during the experiment, F is the residue in experimental boxes (undigested food + excretory products), M is the assimilated food metabolized (calculated by mass balance). All figures are given in mg dry weight. *ECD* is the efficiency of conversion of digested food. High *ECD* values indicate high food conversion efficiency.

One-way ANOVAs, with subsequent Tukey HSD tests were conducted to determine the differences between the various treatments ($P \leq 0.05$).

Results and discussion

Waste reduction efficiency

A food supply of 12.5, 25, 50, and 100 mg day⁻¹ larvae⁻¹ led to a 37.3–43.2% material reduction (dry weight) (Table 2). Supplying 200 mg resulted in a 26.2% reduction. Class 100 showed

the highest *WRI* value, indicating that 100 mg of food per larva and day is the most suitable feeding rate to efficiently reduce waste. Material reduction identified in the present study coincided well with the reported 39–56% cow manure reduction (NC State University 2006) and the 50% chicken manure reduction (Sheppard *et al.* 1994).

Between 17.1% (class 200) and 32.6% (class 50) of the material fed to the larvae was metabolized (Figure 1). Only a small fraction (6.0–16.1%) was transformed into prepupal biomass. The remaining part (55.9–76.9%) was left as residual matter, which had either passed through the gut of the larvae or had not been eaten at all. The efficiency in converting digested food (*ECD*) averaged 29.6% (SE 1.48, range 24.4–38.0%) (Table 2). The *ECD* value indicates how efficient *H. illucens* converts food into biomass. Previous studies on quantitative food processing by insects unfortunately do not provide data on *H. illucens* (Slansky & Scriber 1982, Wiegert & Petersen 1983). To quantify more accurately the food conversion efficiency of the black soldier fly, its assimilation efficiency (AD) is required (Scriber & Slansky 1981). This factor indicates the digestibility of a specific food by a certain organ-

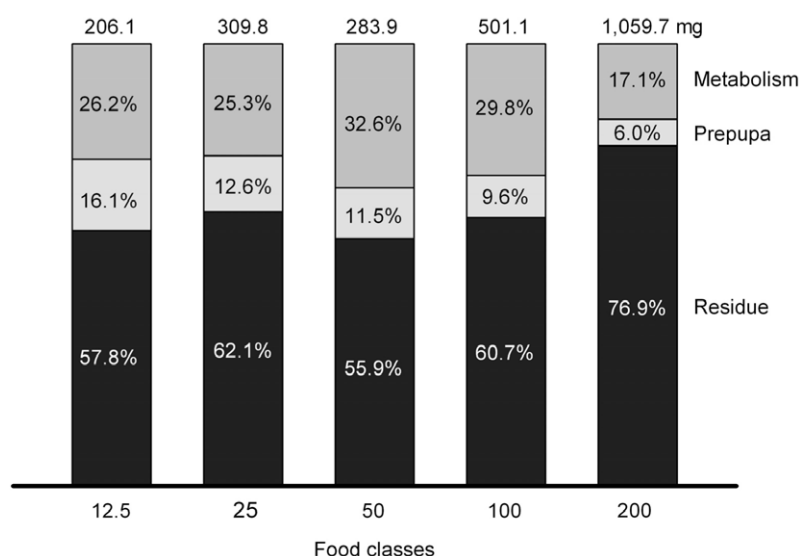


Fig. 1: The relative proportion and the absolute amount of food (five food classes: mg dry weight) that is converted into biomass (prepupal weight, mg in top row), used for metabolism, and remains as residuals. Metabolism is calculated as the difference between residue and prepupal weight and the total amount of food provided during the entire development time.

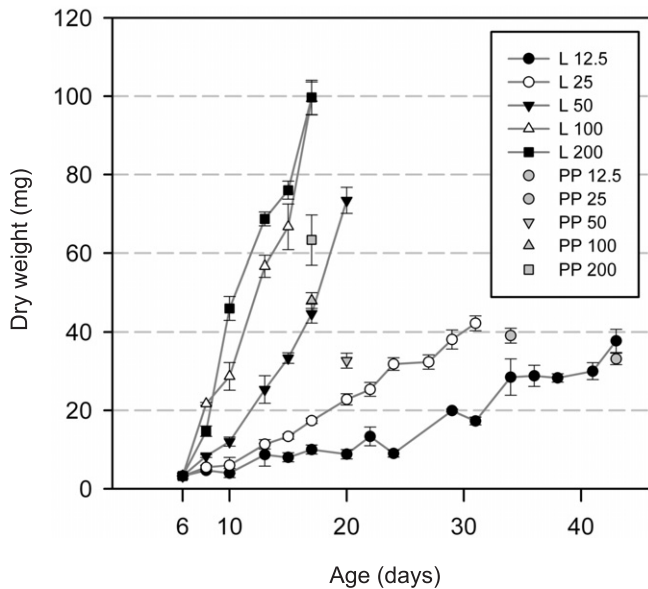


Fig. 2: Increase in dry weight (mean $\pm$ SE) of *H. illucens* larvae with development time, and final weight of the prepupae. Five different feeding treatments: 12.5 to 200 mg chicken feed larva⁻¹ day⁻¹ (60% moisture). L 12.5–L 200, dry weight of larvae; PP 12.5–PP 200, dry weight of prepupae.

ism. However, AD has not been determined in this study. Nevertheless, estimates can be used for *H. illucens* feeding on different waste sources, which can then be used further to adjust operation of a CORS facility.

We should mention that the food added to the feeding boxes may have lost water between individual feeding events. We can, however, not clearly determine to what extent this phenomenon has influenced our results. It is likely that especially the small food rations experienced a stronger dehydration due to the unfavourable surface area-to-volume ratio, and that the drier food could subsequently not be converted to the same extent than under constant moisture conditions.

Weight gain and development time

Larvae fed 200 mg chicken feed (wet weight, 60% moisture) per day revealed the highest prepupal dry weight (mean 63.3 mg, SE 6.36). These larvae required 15.9 days (SE 0.02) development time. Larvae fed with 100 mg day⁻¹ exhibited a prepupal weight of 48.0 mg (SE 2.00) and a development time of 16.6 days (SE 0.31). Food supply of less than 100 mg day⁻¹ resulted in a similar prepupal weight but in a longer larval development time (Table 2 and Figure 2). From the life-cycle viewpoint, the feeding rate of 100 mg food larva⁻¹ day⁻¹ marks the threshold where additional daily food supply does no longer accelerate larval development time (Figure 3).

In the event of food shortage, the larvae are capable of prolonging their development time. This biological trait may simplify operation of a waste management system, as larvae can cope with varying food supply. The lower threshold, where lack of food eventually leads to the death of the larva, was not determined in this study.

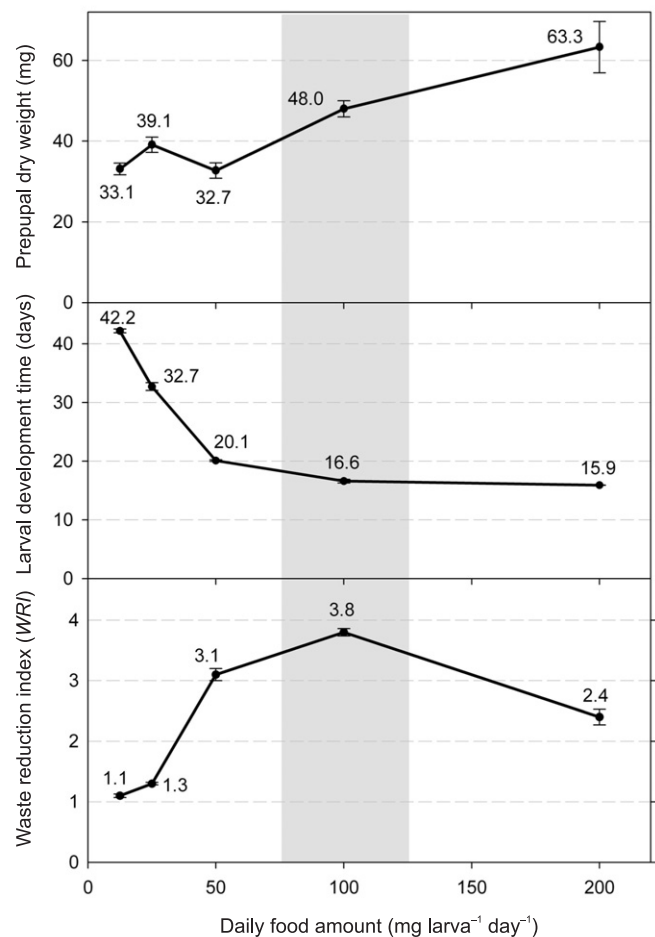


Fig. 3: Prepupal dry weight (mg) larval development time (days) and Waste Reduction Index (WRI), dependent on daily food quantity. Grey area indicates optimal feeding rate.

Prepupal dry weight was similar among classes 12.5, 25 and 50 but differed from class 200 (Table 3). The results show that larvae provided with limited food migrate out of the waste as soon as their prepupal weight reaches ~ 35 mg DW. It is concluded that in the event of food shortage, larvae will feed until they have achieved the minimum energy reserve required to perform pupal development. This critical weight is defined as the minimal weight at which further feeding and growth are not required for a normal metamorphosis and pupation (Nijhout & Williams 1974). There is a time lag from achieving critical weight to secretion of the prothoracicotrophic hormone (PTTH), which causes the larva to stop feeding. The weight gain within this period may explain the differences in prepupal weight between the various feeding types.

As expected, adults that underwent food stress during their larval development (classes 12.5, 25 and 50) showed reduced dry weights in comparison with the other two classes (Table 3). Adult females weighed 1.3–1.5 times more than males, typically for arthropods.

Protein content

The relative protein content of prepupae ranged from 31.9% (SE 1.74, class 200) to 46.3% (SE 1.88, class 12.5) (Table 4).

Table 3: Prepupal and adult weight (mg) of *Hermetia illucens* fed with five different amounts of chicken food (12.5, 25, 50, 100, and 200 mg larva⁻¹ day⁻¹, 60% moisture).

		12.5 mg		25 mg		50 mg		100 mg		200 mg	
		Mean	SE	Mean	SE	Mean	SE	Mean	SE	Mean	SE
Prepupae											
Wet weight	mg	88.5a	4.47	103.0ab	5.64	98.7a	3.84	121.3b	2.40	157.3c	8.19
Dry weight	mg	33.1ab	1.44	39.1ab	1.90	32.7a	1.92	48.0b	2.00	63.3c	6.36
Adults											
Male, dry weight	mg	12.7a	0.48	16.8b	1.69	16.6b	0.24	24.8c	0.66	29.7d	0.70
Female, dry weight	mg	17.9a	0.43	24.4b	1.17	25.1b	0.30	33.5c	1.89	37.7c	0.72

Mean values followed by the same letter in the same row do not vary significantly ($P < 0.05$).

Table 4: Larval, prepupal and adult protein content (% of dry weight) of *Hermetia illucens* fed with different amounts of chicken food daily (12.5, 25, 50, 100, and 200 mg larva⁻¹ day⁻¹, 60% moisture). Protein calculation is based on total nitrogen $\times 6.25$, except for prepupae (chitin-corrected values, see text).

		12.5 mg		25 mg		50 mg		100 mg		200 mg	
		Mean	SE	Mean	SE	Mean	SE	Mean	SE	Mean	SE
Larvae	%	42.4a	1.29	38.9ab	0.84	38.6ab	1.53	37.3ab	1.27	34.9b	0.97
Prepupae											
Total	%	46.3a	1.88	36.4b	1.35	36.6b	0.33	38.1b	0.61	31.9b	1.74
Chitin corrected	%	42.5a	1.88	32.6b	1.35	32.8b	0.33	34.4b	0.61	28.2b	1.74
Adults											
Male	%	52.9a	0.71	48.7a	2.31	51.4a	0.23	49.0a	0.88	47.9a	0.59
Female	%	52.4a	0.93	47.7b	1.10	51.8ac	0.71	49.0abc	0.83	48.4bc	0.58

Mean values followed by the same letter in the same row do not vary significantly ($P < 0.05$).

The chitinous fraction of prepupae was 8.72% DW (SE 0.71). Hence, the chitin-corrected protein contents ranged from 28.2% (SE 1.74, class 200) to 42.5% (SE 1.88, class 12.5).

The protein contents of the well nourished prepupae obtained in this study turned out to be slightly lower than the 42–44% observed by Booram *et al.* (1977), Hale (1973) and St-Hilaire *et al.* (2007a). Nevertheless, the protein content obtained approves the applicability of dried prepupal meal in the feed industry.

Conclusions

It was concluded that a daily rate of 100 mg chicken feed per larva met both the demand for a nutrient-rich prepupal output and high organic matter degradation within a short time (Figure 3).

Using the digestible energy as a reference value, the present results can be projected to different kinds of organic waste (Table 5). Depending on larval density and waste source, a daily load of 3–8 kg m⁻² can be applied to a black soldier fly CORS system. Needless to say that the calculated feeding rates presented in Table 5 are based on rough estimations and have to be interpreted carefully as they depend on other factors such as material moisture or fibre content.

Assuming a larval density of 5 larvae cm⁻², a prepupal dry mass of 2.5 kg could be harvested per life cycle and m² (145 g m⁻² day⁻¹) when fed with a waste equivalent of 100 mg chicken feed larva⁻¹ day⁻¹.

For farmers and small entrepreneurs in low- and middle-income countries, biomass produced by black soldier flies can be an additional income derived from this low-tech based waste treatment. It has been shown that even an irregular waste input (e.g. due to seasonal or personal fluctuations) the larva population remains alive albeit with reduced metabolic activity until the next food/waste input.

This study has taken a major step towards the quantification of a black soldier fly CORS system. However, our results were obtained from small-scale laboratory experiments. Implementation at a larger scale *in situ* will lead to new challenges, especially under changing environmental conditions (e.g. humidity, incident solar radiation, predators/parasitoids), the design of the treatment system (operability and usability), and the fate of problematic waste contents (e.g. heavy metals, pathogens) will be new elements to tackle.

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Table 5: Projected daily waste loading rates for a *Hermetia illucens* waste management system based on the energy values of different organic waste sources: calculation basis: 100 mg chicken feed larva⁻¹ day⁻¹ (60% moisture).

	Gross energy (MJ kg ⁻¹)	Digestible energy (MJ kg ⁻¹)	Daily feeding rate (mg larva ⁻¹ day ⁻¹)	Waste loading rate/m ² (kg) ^a	Energy data source
Chicken feed ^b	–	11.7	100	5.00	
Kitchen waste	22.0	19.3 ^c	61	3.05	(ALP 2007)
Vegetable waste	17.6	11.9 ^c	98	4.90	(ALP 2007)
Green banana	17.8	11.4 ^d	103	5.15	(Ulloa <i>et al.</i> 2004)
Pig manure	17.8	7.4 ^d	158	7.90	(Ulloa <i>et al.</i> 2004)
Poultry manure	14.6	6.7 ^d	175	8.75	(Ulloa <i>et al.</i> 2004)
Human faeces	21.6	–	130 ^e	6.50	(Diem & Lentner 1968)

^aAssumption: 50 000 larvae m².

^bUFA 625 Alleinfutter für Legehennen UNIVERSAL.

^cDigestible energy for pigs.

^dPotential digestible energy for fish, calculated according to digestible energy coefficients for *Ictalurus punctatus* (channel catfish).

^eCalculation assumed the same gross energy/digestible energy ratio as for pig manure.

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